CIRCADIAN INFLUENCE ON ATHLETIC PERFORMANCE

Focus on Baseball

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INTRODUCTION

Circadian Aspects

Since the beginning of humankind, our body functions, as well as our physical and mental performance abilities have become oriented and governed by circadian (i.e. 24-hour) rhythms. Think of it as a very sophisticated time of day "wiring" system that controls virtually all of our biological and physiological functioning. This system is illustrated in Figure 1. Around 23:00 we go to bed. At midnight we are having the first episode of deep sleep. We receive a dose of growth hormone in the first several hours of sleep to repair muscle and tissue damage. Our immune system is working at its highest level at this time. We are sleeping in recurring cycles of approximately 90 minutes.[1] Our minimum core body temperature is reached between 03:00 and 05:00. This is the time of the highest sensitivity to light. To maintain high athletic performance during the day, and a stable bio-rhythmus, one should be sleeping at this time and avoiding any light exposure. We call the important sleep during this time "anchor sleep". Avoiding any light between 03:00 and 05:00 keeps the circadian rhythm strong, stable and in sync. Starting around 04:00 blood pressure and cortisol (stress hormone) levels increase and melatonin disappears. After 8 hours in bed and 7.5 hours of sleep we get up around 7:00. After breakfast we reach a peak in mental alertness around 10:00. This would be a good time to talk about game strategy and other important game relevant aspects that require thought processing.

After lunch, our physiology falls into the so called "post lunch dip", or afternoon low. This time could be a good opportunity for a 15 minute nap. Starting around 14:00, the best time for baseball begins. Between 14:00 and 19:00 our physical abilities, are elevated in terms of coordination, reaction time, cardiovascular efficiency, muscle strength, and physical performance. These are obviously all important factors in a baseball game, and they occur during this time window of high core body temperature. Unfortunately, only a minority of the baseball games fall in this time slot. The majority of games start in the evening around at 19:00 and last approximately 3 ½ hours (yellow area in Figure 1). To be at peak performance between 19:00 and 22:00, circadian "phase delay" approaches need to be applied for optimal game preparation strategy and improved performance results.

Before this is discussed, it is important to first understand the baseball game schedule and the impact it has on each team. Game data from the 2012 season was analyzed for this purpose, since it provided the most recent win/loss performance results. The playoff games of 2012 are excluded, since the playoff situation is quite different from the regular season, and the "circadian advantage" strategies would need to be adjusted accordingly. Rather, this paper is about managing the regular season to increase the probability of making the playoffs.

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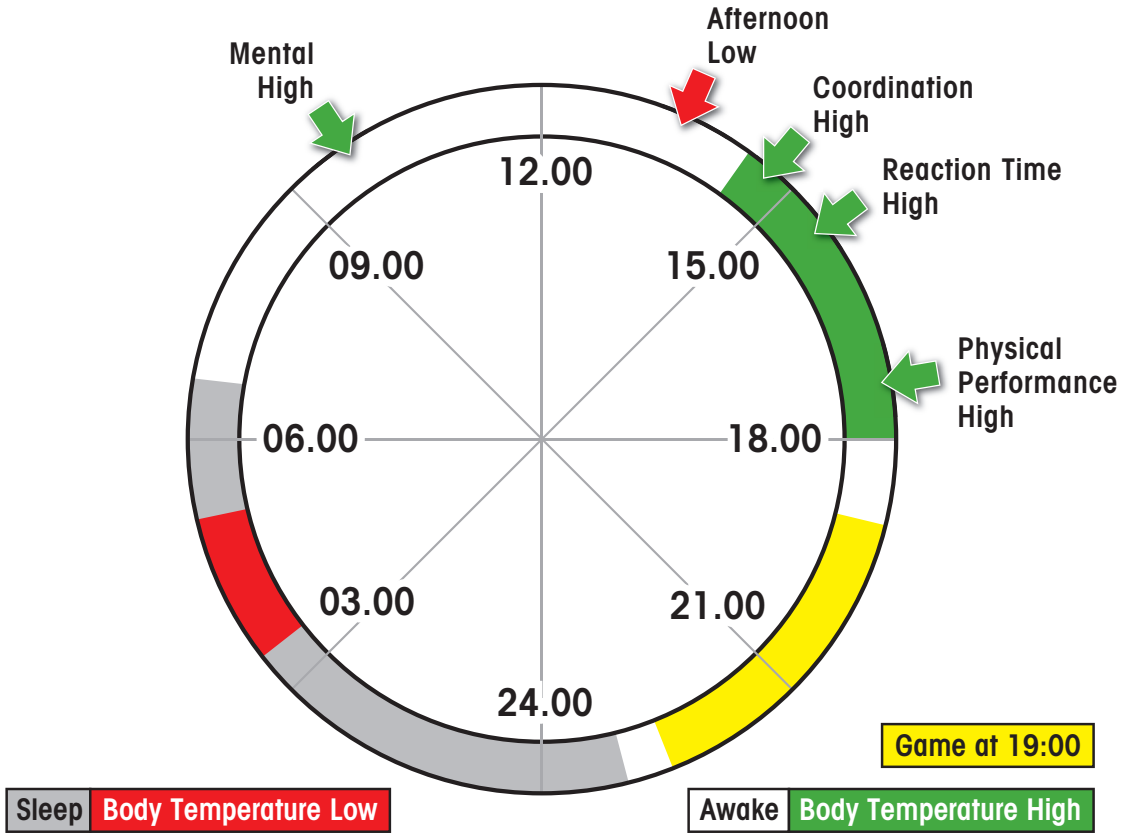
Baseball Aspects

To say the least, the game schedules in Major League Baseball are very demanding, not only for the teams and their organizations, but especially for the players. The season starts in April and ends with the World Series in October. Each team plays 162 games in approximately 180 days with only 2 or 3 days off per month. This means, on average, about one game every day with frequently changing game start times and varying game locations. The games start at different hours of the day, like noon, 13:00, 14:00, 15:00, 18:00, 19:00, 20:00, 21:00, and 22:00 (EST). These varying time conditions make it very challenging for players, coaches and trainers to optimally prepare for the game and achieve peak performance throughout its duration. Unfortunately, **this constant shifting of game times, game locations, and travel schedules desynchronizes the circadian timing mechanism to create a type of jet lag that can be referred to as "game lag".** Perhaps this explains some of the listless and sluggish performances that every team experiences from time to time.

FIGURE 1:
Circadian Body Clock: Important Times for Athletic Mental and Physical Performance.



Phase delaying strategies will shift peak performance to later times, and phase advancing will shift it to earlier game times.

Thus, one important reason for the difficulties in game preparation and performance is the human circadian system. Our biological clocks exhibit different lows and highs in coordination, reaction time, and physical strength as illustrated above. The physiological stress imposed by having to adjust to each of the different game starting times and locations creates disturbances in our biological cycles and weakens the circadian rhythms for the players and coaches. This disturbance will be amplified by traveling to other time zones. **Therefore, it would seem to make sense to adapt circadian strategies to the game times when most of the games are scheduled, and to stick to it.** The purpose of this study was to determine if such a strategy would, in fact, positively influence a Team's success, and we've defined success as "making the playoffs". To determine this also required a common metric to track the results, and thus we introduced the abbreviation **"TWoMFS"** for **Time Window of the Most Frequent Starts**. Our hypothesis is that teams who have a high number of games in the TWoMFS, (i.e. with games that start within the same one hour time window) should have an inherent "Circadian Advantage". To demonstrate this point, we show the 2012 game start time distribution for two teams. The first team (Colorado Rockies, Figure 2a) displays a relatively broad distribution of game start times, and a peak of 45 games in the **Time Window of the Most Frequent Starts (TWoMFS)** scheduled at 21:00 EST. The second team (Baltimore Orioles, Figure 2b) displays a more concentrated distribution of game start times, with a peak of 91 games scheduled in the **TWoMFS** of 19:00 EST. Looking at Figure 2a and Figure 2b from a circadian advantage viewpoint, the Baltimore Orioles had a clearly built in circadian advantage of 46 games (91-45 TWoMFS) in 2012 compared to the Rockies.

FIGURE 2A:
Maximum Number of Games (45) at 21:00 EST in the Time Window of Most Frequent Starts (TWoMFS).

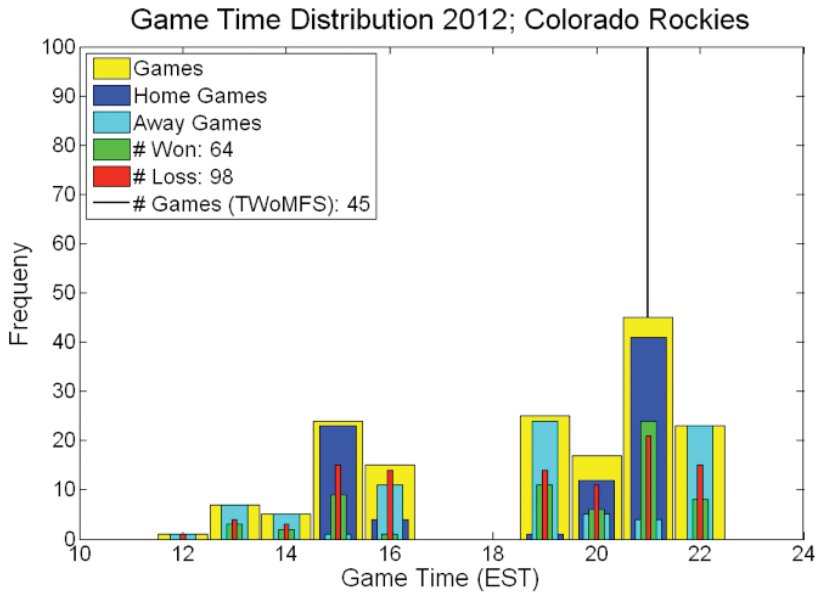
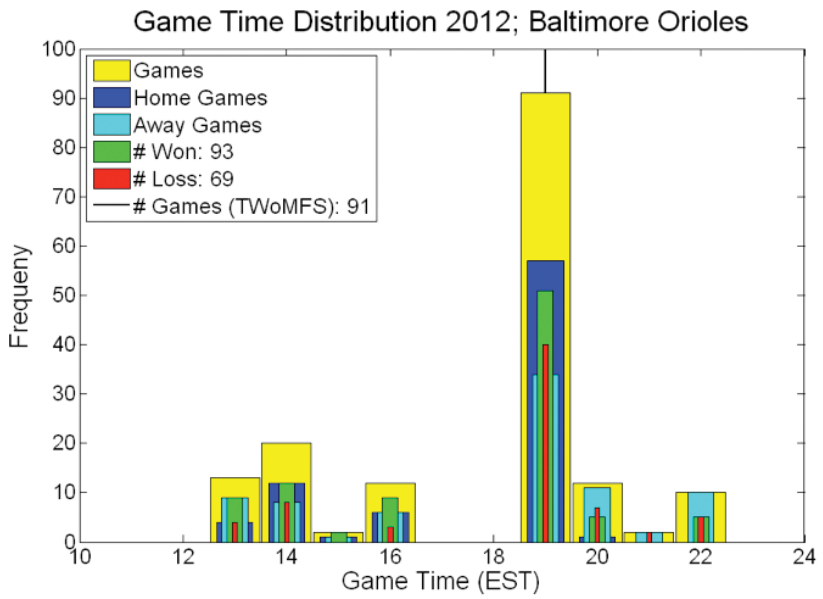


FIGURE 2B:
Maximum Number of Games (91) at 19:00 EST in the Time Window of Most Frequent Starts (TWoMFS).



Each year, every team has a specific **TWoMFS** with a dramatically different number of games in that "Wheel House" of opportunity. To increase their chances of making the playoffs, teams with a high number of games in the TWoMFS would have to win most of these games. Thus, we postulate that focusing on those games that fall in their TWoMFS would be the most basic and simplest strategy for gaining a "Circadian Advantage".

To further demonstrate the impact that the game schedules could have on the ability of the teams to adjust to the changing game times, we divided the teams in two groups: The Central/West Coast group (Table A), and The East Coast group (Table B). The Central/West Coast teams have only an average of 61.5 games (Table A) scheduled in their TWoMFS. The East Coast teams, on the other hand, have an average 81.3 games (Table B) in the TWoMFS. It turns out that the difference of approximately 20 games between the two groups is highly significant based on the 2012 won-lost results. Indeed, the difference correlated with the average number of games won by each group. The East Coast teams (Table B) won an average of 4.5 more games (83.4-78.9) in 2012 than the Central/West Coast teams (Table A). Moreover, a comparison of the percentage of overall games won shows an average advantage of 51.7% to 48.7% in favor of the East Coast teams.

This suggests that the average number of games scheduled and won in the TWoMFS window can be a predictor for a winning or losing season. Moreover, all of the teams who made the playoffs in both leagues had relatively higher numbers of winning games in the TWoMFS (Table A

and B), as well as a higher percentage of wins in that window compared to the overall average. This further suggests that the number of games scheduled in the TWoMFS window may not only be a predictor for a winning season, but also for making the playoffs! Better yet, there is a clear opportunity for leveraging the win margin even further through the application of circadian game preparation strategies that follow.

Given today's payroll costs and the additional playoff revenues at stake, an extra 4-5 or more wins could translate into millions of dollars on a team's bottom line…not to mention increased brand value and bragging rights.

TABLE A:
2012 Regular Season Statistics for the Central-West coast Teams.

Teams	Won	Loss	# Games (TWoMFS)	W (TWoMFS)	L (TWoMFS)	Won [%]	Won [%](TWoMFS)
Arizona Diamondbacks	81	81	69	37	32	50	54
Los Angeles Angels	89	73	67	41	26	55	61
Los Angeles Dodgers	86	76	71	29	42	53	41
Oakland Athletics	94	68	56	36	20	58	64
San Diego Padres	76	86	61	32	29	47	52
San Francisco Giants	94	68	61	35	26	58	57
Seattle Mariners	75	87	63	29	34	46	46
Colorado Rockies	64	98	45	24	21	40	53
Chicago Cubs	61	101	51	28	23	38	55
Chicago Whitesox	85	77	63	33	30	52	52
Houston Astros	55	107	70	24	46	34	34
Kansas City Royals	72	90	65	36	29	44	55
Milwaukee Brewers	83	79	60	36	24	51	60
Minnesota Twins	66	96	59	25	34	41	42
St. Louis Cardinals	88	74	59	34	25	54	58
Texas Rangers	93	69	64	39	25	57	61
Avg. Central-West	**78.9**	**83.1**	**61.5**	**32.4**	**29.1**	**48.7**	**52.9**

Playoff Teams	Won	Loss	# Games (TWoMFS)	W (TWoMFS)	L (TWoMFS)	Won [%]	Won [%](TWoMFS)
Los Angeles Angels	89	73	67	41	26	55	61
Oakland Athletics	94	68	56	36	20	58	64
San Francisco Giants	94	68	61	35	26	58	57
St. Louis Cardinals	88	74	59	34	25	54	58
Texas Rangers	93	69	64	39	25	57	61
Avg. Central-West	**91.6**	**70.4**	**61.4**	**37.0**	**24.4**	**56.5**	**60.3**

Teams marked in green made the playoffs.

TABLE B:
2012 Regular Season Statistics for the East Coast Teams.

Teams	Won	Loss	# Games (TWoMFS)	W (TWoMFS)	L (TWoMFS)	Won [%]	Won [%](TWoMFS)
Atlanta Braves	94	68	77	42	35	58	55
Baltimore Orioles	93	69	91	51	40	57	56
Boston Redsox	69	93	88	37	51	43	42
Cincinnati Reds	97	65	74	40	34	60	54
Cleveland Indians	68	94	73	31	42	42	42
Detroit Tigers	88	74	64	34	30	54	53
Miami Marlins	69	83	87	34	53	45	39
New York Mets	74	88	78	31	47	46	40
New York Yankees	95	67	86	53	33	59	62
Philadelphia Phillies	81	81	84	42	42	50	50
Pittsburgh Pirates	79	83	85	41	44	49	48
Tampa Bay Rays	90	72	88	50	38	56	57
Toronto Bluejays	73	89	81	36	45	45	44
Washington Nationals	98	64	82	53	29	60	65
Average East	**83.4**	**77.9**	**81.3**	**41.1**	**40.2**	**51.7**	**50.5**

Playoff Teams	Won	Loss	# Games (TWoMFS)	W (TWoMFS)	L (TWoMFS)	Won [%]	Won [%](TWoMFS)
Atlanta Braves	94	68	77	42	35	58	55
Baltimore Orioles	93	69	91	51	40	57	56
Cincinnati Reds	97	65	74	40	34	60	54
New York Yankees	95	67	86	53	33	59	62
Washington Nationals	98	64	82	53	29	60	65
Average East	**95.4**	**66.6**	**82.0**	**47.8**	**34.2**	**58.9**	**58.2**

Teams marked in green made the playoffs.

Note: Teams are aligned geographically…not by conference or league.

Circadian Game Preparation

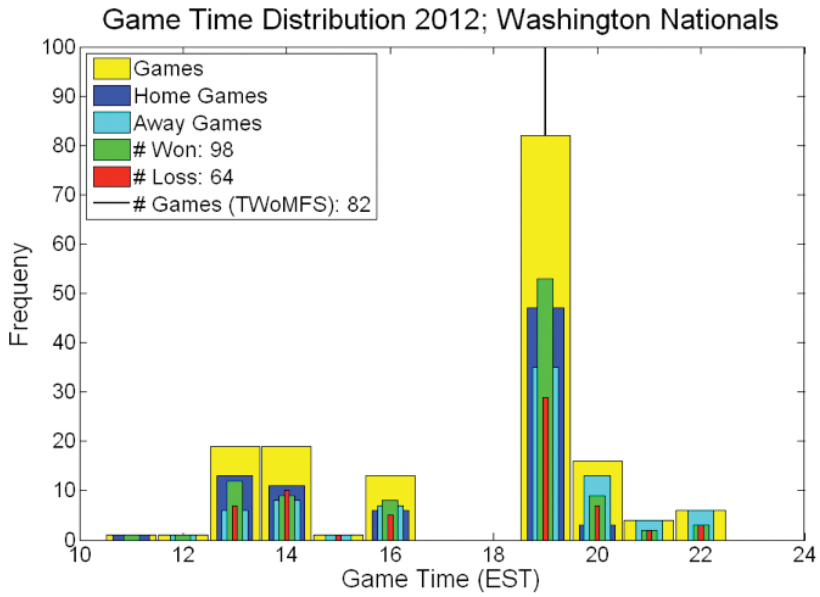
Based on this study of the 2012 season schedule and win/loss results, **there is a strong indication that a key to achieving more wins may be how well a team focuses on their TWoMFS, and makes it a priority to win these games.** The right circadian preparation strategy can't win games alone. Less talented and/or less healthy teams will lose no matter what. But the circadian factor can provide an advantage in games between comparable teams. Experience and scientific research shows that athletic performance in all sports[3] can be improved by approximately 10% with the right circadian preparation strategy. This could well make the difference between winning and losing close games, between making or missing the playoffs, and between team profit and loss.

The East Coast Strategy:

For all East Coast teams the most frequent game start time in 2012 was 19:00 EST, or within the 1-hour time window of most frequent starts between 18:30 and 19:30 (EST). A recommended preparation strategy under this circumstance would simply be to phase delay the collective "body clock" of the team by 2-3 hours in order to achieve maximum biological and physiological performance during those games in this Circadian Advantage window. **The enhanced ability to win over 60% of the games in the TWoMFS may very likely get a team into the playoffs.** In the 2012 season this was evident with the New York Yankees (62%), and Washington Nationals (65%). This concept is straightforward and minimizes more complicated circadian advance and delay strategies. The body clock of the players could be tuned to the predominant 19:00 EST game times, during training camp or early in the season, for example, in preparation for the 2013 season. How to delay the body clock by approximately 2-3 hours is not so complicated. It simply means going to bed later and getting up later than indicated in Figure 1 to shift performance capability to the later game times. Players should not get up before 08:00, but then they should take a nap around 14:30 for a maximum of 20 minutes. After the nap, players should be outside preparing for the game as usual and getting exposure to bright (outdoor) light. **The success of the strategy depends on establishing a strong circadian rhythm with the players.** To achieve this, special attention to the anchor sleep time between 03:00 and 05:00 is also required.

If the players have successfully delayed their body clocks by staying up later each night (e.g. midnight) and sleeping in later (e.g. 08:00), they have to stick to this sleep-wake schedule each day during the season. It is important to keep the body clock tuned to the 19:00 games, (or whatever their TWoMFS might be), even in preparation for afternoon game starts (e.g. between 12:00 and 15:00 EST). The team will still win their share of the early games, but they will not disturb or weaken their established body clock. As a result, they will win more games in the circadian time window advantage in their playing schedule. A case study example for the successful application of the East Coast strategy is illustrated in Figure 3. The Washington Nationals won 53 and lost only 29 of their (2012) 19:00 EST games. This is a winning percentage of 65% in the 19:00 "Circadian Advantage" TWoMFS, and certainly contributed to their break-out year.

FIGURE 3:
Most games won (53) in the Time Window of most Frequent Starts at 19.00 EST.

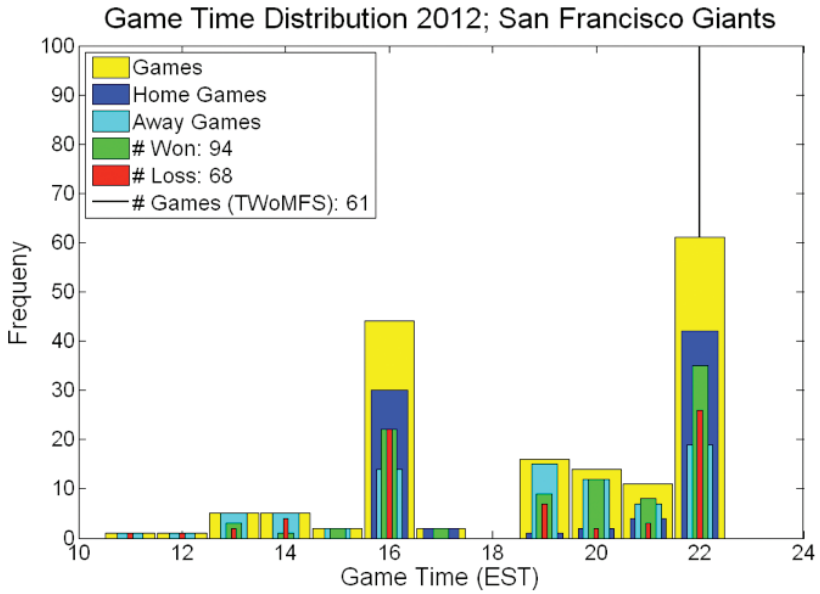


The Central - West Coast Strategy:

For the Central/West Coast teams the situation is a bit more challenging. They had considerably fewer game starts in their TWoMFS compared to the East Coast teams, as described above. Still, achieving above a 60% winning rate in their TWoMFS was also a good predictor for the 2012 playoffs. The Los Angeles Angels with 61% a winning score in the TWoMFS, the Oakland Athletics with 64%, and the Texas Rangers with 61% all made it to the playoffs in 2012.

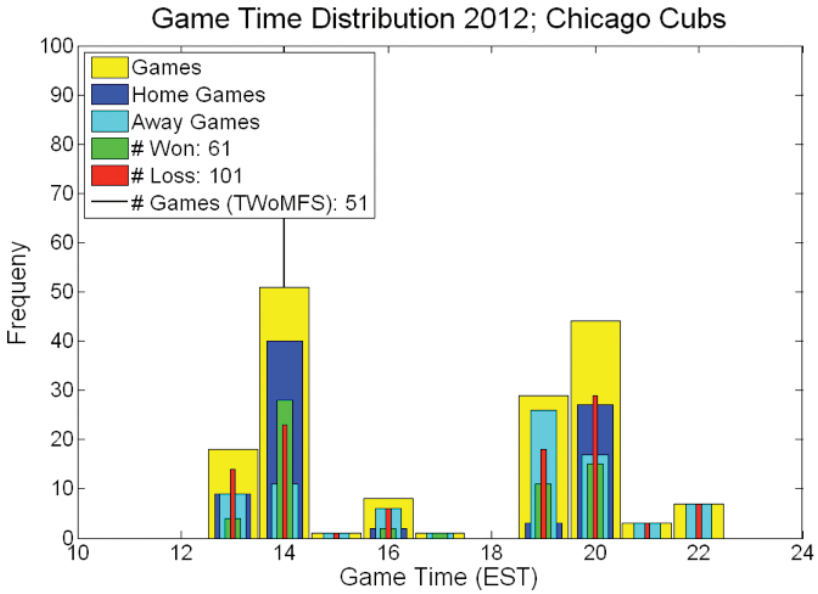
In general, however, to simply tune the team body clocks to a single start time might not be sufficient for the Central-West Coast strategy to win enough games to make the playoffs. A more complex approach may be needed, including having to apply both circadian phase advance and phase delay strategies. The most successful team in 2012, the San Francisco Giants, won the majority of their games (64) in the 19:00, 20:00, 21:00, and 22:00 time windows with a winning percentage of 63% (Figure 4). To achieve a successful Central/West Coast strategy, teams will have to adjust their player's body clocks to several time windows to achieve maximum performance. Around the Giants 20:00 game start time, for example, a broader circadian range of adjustment (e.g. 19:00-22:00 EST) can be defined as their TWoMFS focus. Through scientific studies, it is known that the circadian clock can be delayed by 1 hour and advanced by ½ hour per day.[2] The biological clock can also be influenced by zeitgebers (e.g. time givers, such as bright light).[2] Outdoor light in the morning will advance the biological clock, and exposure to light in the evening will delay the biological clock.

FIGURE 4:
High number of games (64) won in broad window between 19:00 and 22:00 EST.



In terms of the most **"Most Challenging Circadian Game Schedule"** for 2012, the "award" has to go to the **Chicago Cubs (Figure 5)**. As pointed out earlier, there are easier and more challenging game schedules from a Circadian Advantage viewpoint. In general, most of the East Coast teams last year benefitted from a predominant 19.00 game start time, enabling them to more readily achieve a won-lost advantage by focusing more on the specific games in their TWoMFS. This was not the case for the Central/West Coast teams, but only the Cubs had a schedule in 2012 where the maximum number of games occurred in an afternoon time window (14:00 EST), with a smaller peak in the evening at 20:00 EST. This created the most challenging schedule to manage in 2012. Nevertheless, the Cubs had a positive winning record of 55% in their 14:00 TWoMFS, but unfortunately, they lost most of the late games. Perhaps the application of more aggressive circadian phase advancing and delaying techniques might have changed this outcome.

FIGURE 5:
Example of the most challenging game schedule in 2012.



The only team with the TWoMFS in the afternoon at 14:00 EST. They also had a secondary peak between 19:00 and 20:00 (EST).

Conclusion

The scientific evidence is clear:[3] Athlete's peak performance in most sporting events involving physical power occurs in the late afternoon or early evening. For example, swimmers in a 100 meter race perform better at 17:00 than at 07:00. Best performances for the 400 meter freestyle were observed in the evening.[3] In cycling, the sprinting power measured by ergometers is greater between 17:00 and 19:00 compared to 07:00 and 09:00. Sprinters achieve the best results over 80 meters at 19:00. Serve velocity of tennis players is greater at 18:00 than at 9:00. There are many more examples, but all have in common the fact that the best time for short term physical performance (e.g. pitching and batting) always coincides with our circadian rhythms, in general, and with high core body temperatures, in particular.

The analysis of the 2012 baseball season from this circadian perspective was no different, and it revealed some unanticipated results. In particular, the East Coast teams were scheduled for more games in the **Time Window of Most Frequent Starts (TWoMFS)** than were the Central and West Coast teams. There was also a clear correlation to both the number of games played in the

TWoMFS, and the percentage of those games that were won. **Moreover, the percentage of games won in the TWoMFS showed strong correlation with making the playoffs.** In fact, six teams (two from the East Coast group and four from the Central-West Coast group) made the playoffs with a winning score over 60% in the TWoMFS! These percentages were obviously even more significant for those teams that had the highest number of games scheduled in a single time window.

Each team schedule certainly brings some specific circadian performance challenges, but also presents new opportunities for trainers and coaches. By similarly analyzing the game schedule(s) for the 2013 season, more simple strategies and thus game winning performance opportunities can be developed to mitigate "game lag" disruption during the incoming season. Some groundwork can be laid during the current spring training. The 2013 schedule can be analyzed to determine a team's TWoMFS. To the degree possible, practices should be scheduled at the same time of day that peak performance is required for that 2013 TWoMFS. Phase delaying and advancing strategies can be developed for managing the TWoMFS games and for long road trips over multiple time zones. By applying the scientifically established principles of circadian physiology, these scheduling challenges can be mitigated to increase competitiveness and performance results on any given day. By focusing on the most prevalent game time windows, and building a strong team body clock pre-disposition to those windows through phase delaying and advancing strategies, the chances for making the playoffs can be significantly improved.

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